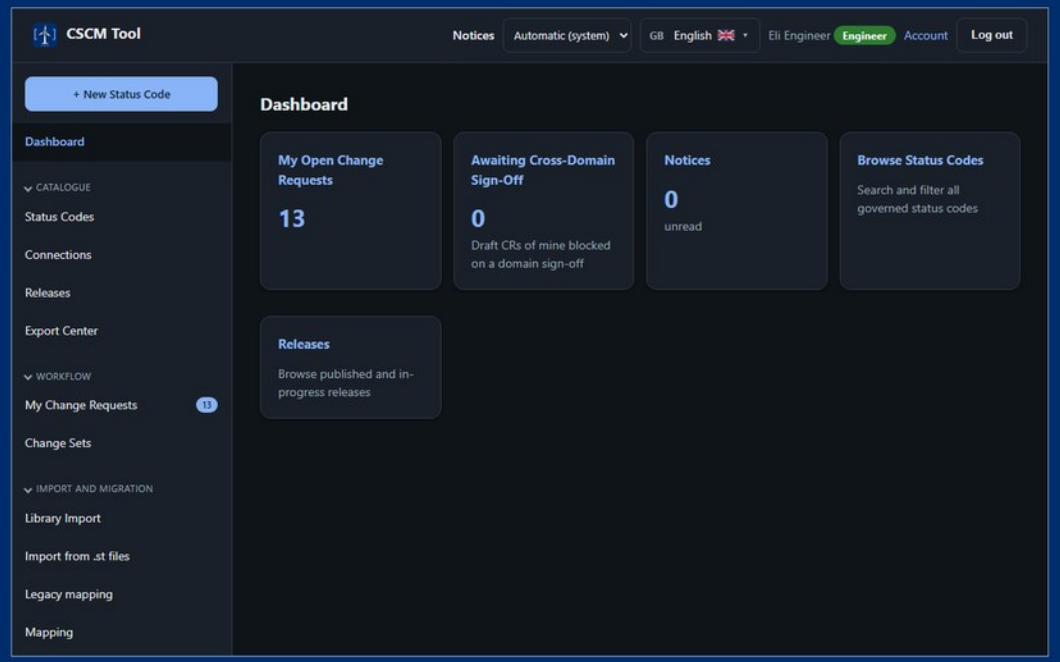
Software Owner and Administrator: Bastian Ritter · bastian.ritter@senvion.in

# CSCM Tool

## Controller Status Code Management Tool

## Management overview




Main Controllers and Hub Controllers of wind turbines

Version 4.18.1

2026-10-29

# One governed source of truth for every status code

### Before

Codes lived in documents. A change was hard to trace and hard to reuse.

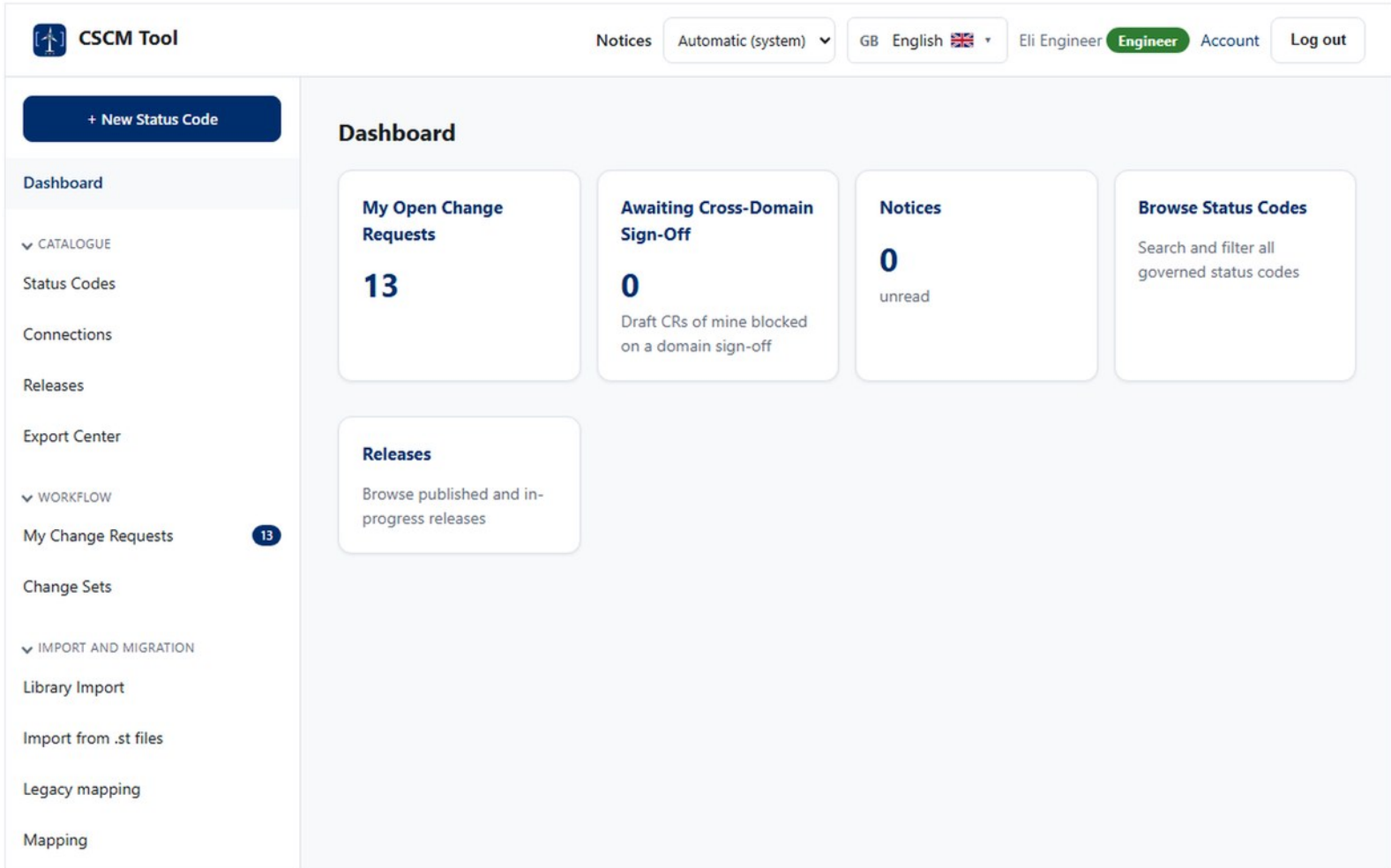
### With the tool

A code is created, reviewed, approved, versioned and released in one place.

### The result

Manuals, specifications and SCADA lists are produced from approved codes only.

# Everything people need, in one place




1 **A menu for the role**
Each person sees only what their role may do.

2 **Work that waits**
Counters and links lead straight to reviews, approvals and drafts.

3 **Colour with meaning**
Roles and lifecycle states have fixed colours, never decoration.

# Three independent decisions before a release




### 1 Reviewer

Checks content and wording, and signs off.

### 2 Administrator

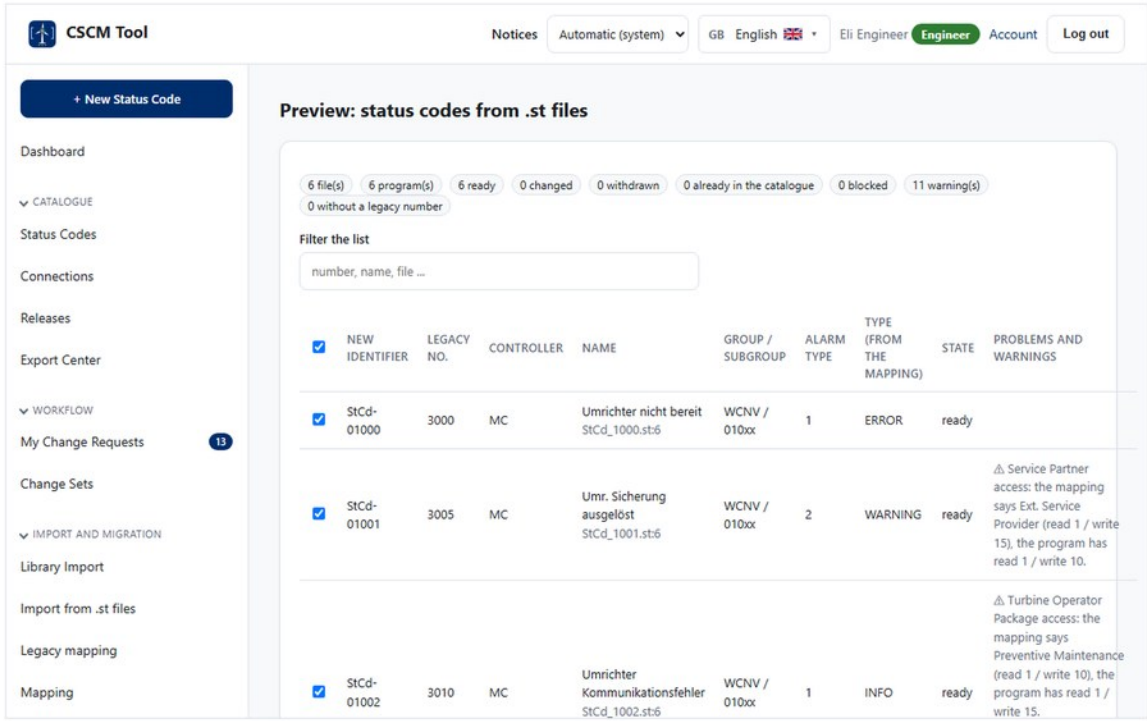
Confirms audience access and signs off.

### 3 Chief Engineer

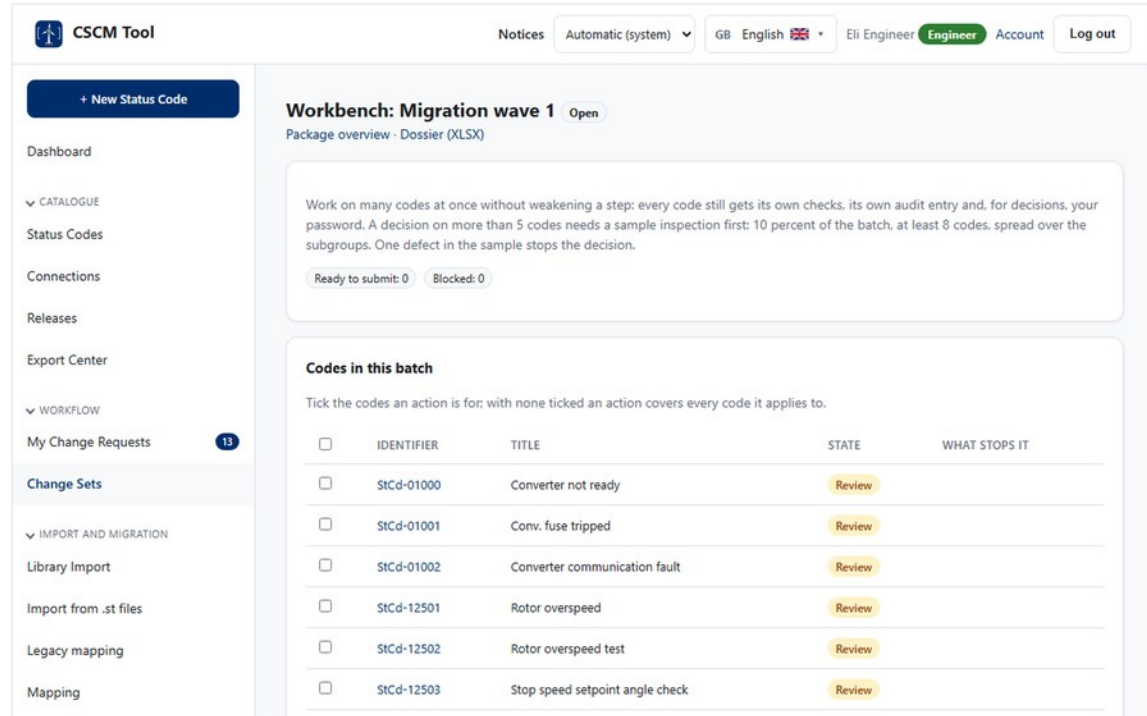
Approves. Nobody can approve their own work.

Before the review, the engineering domains concerned sign off the content (cross-domain sign-off). Every step is recorded in an audit log that cannot be changed.

# The first migration: about 1,250 codes




Programs are read from the controller files and previewed. Nothing is created until a person has looked.




The workbench lets each step work on the whole batch, with a sample inspection that one defect can stop.

# Built to be trusted

**1,280**

automated tests, all passing

**93 %**

of the application code covered

**230**

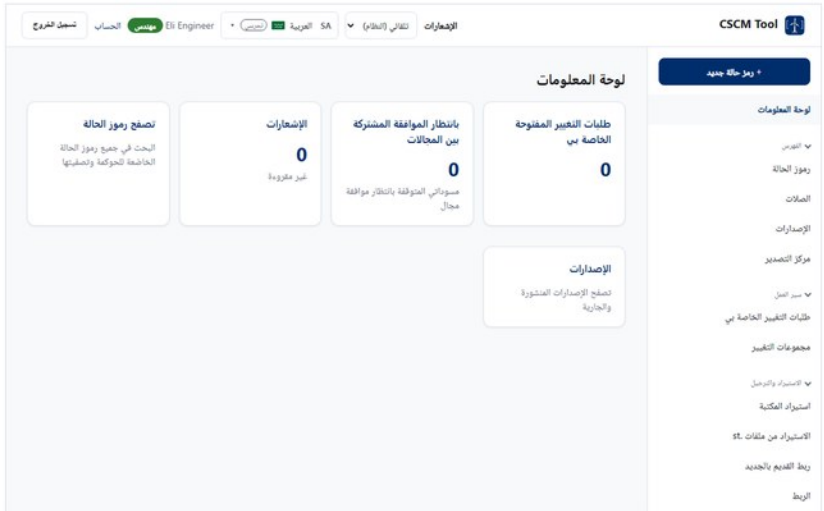
requirements, each traced to tests

**19**

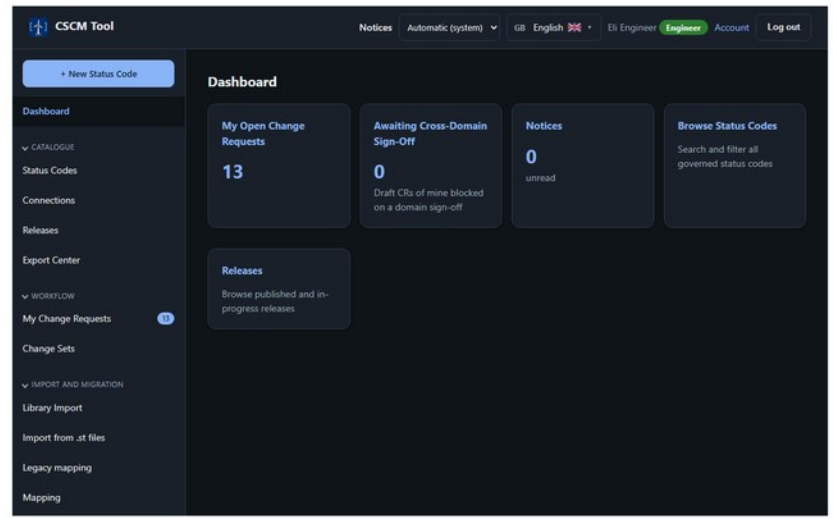
languages, including right to left

The REST API has 171 operations, the database 65 tables. Every change of a code is kept with who, when and why.

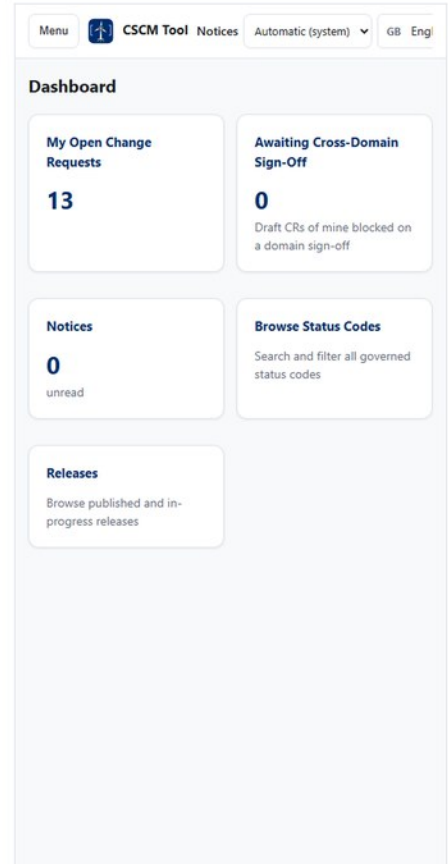
# Looks right in every language and light




Right to left for Arabic

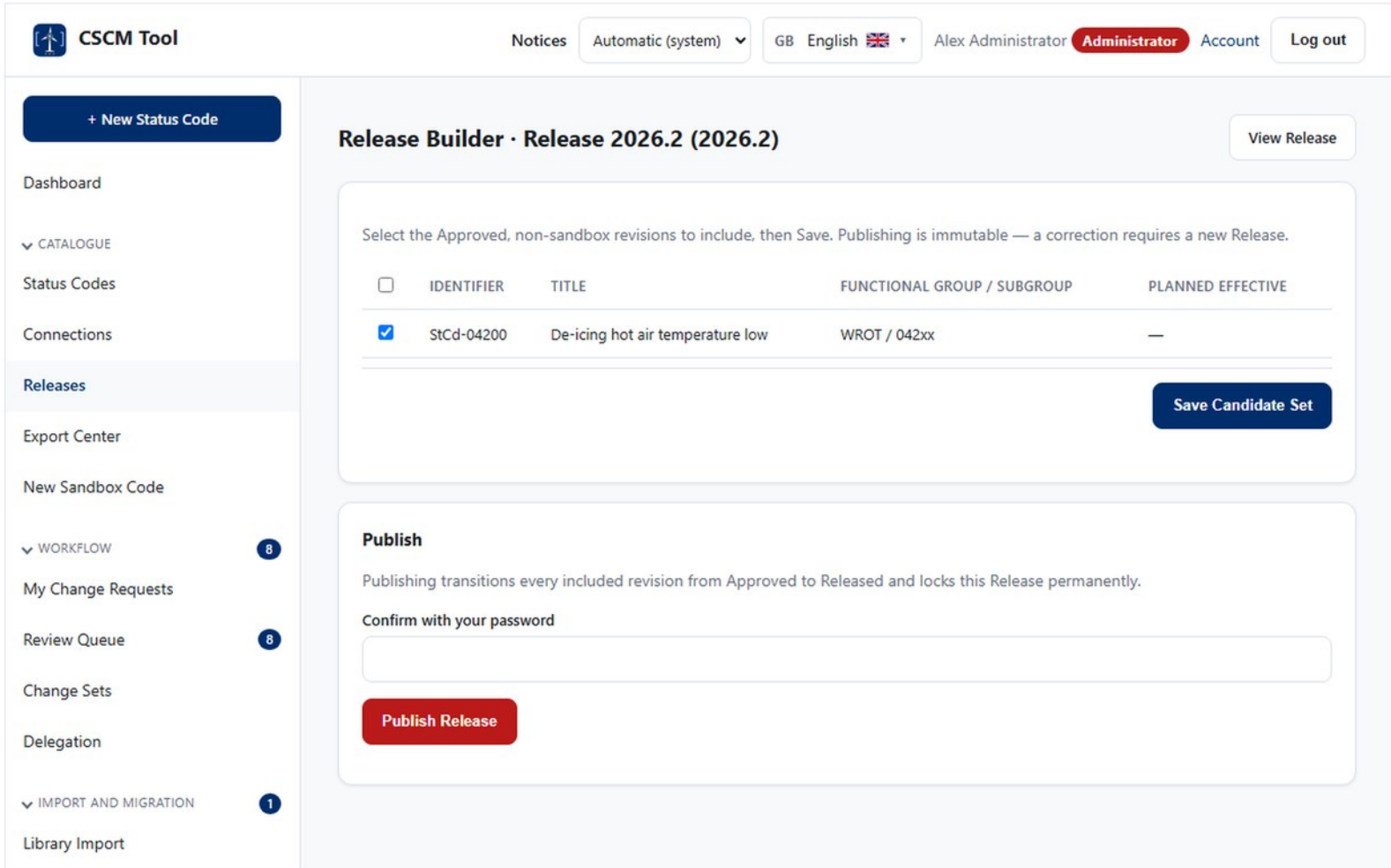



Dark theme, set by the system or by choice




And on a phone

# Releases people can rely on

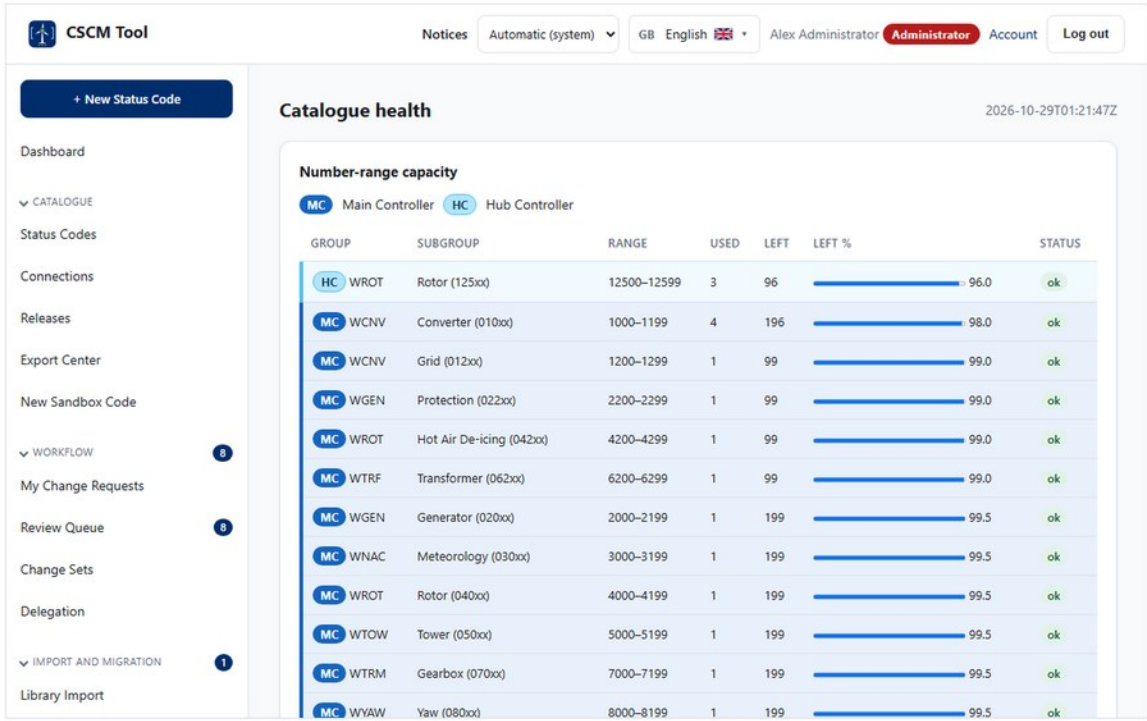



1 **A fixed baseline**
A release holds the latest published revision of every code.

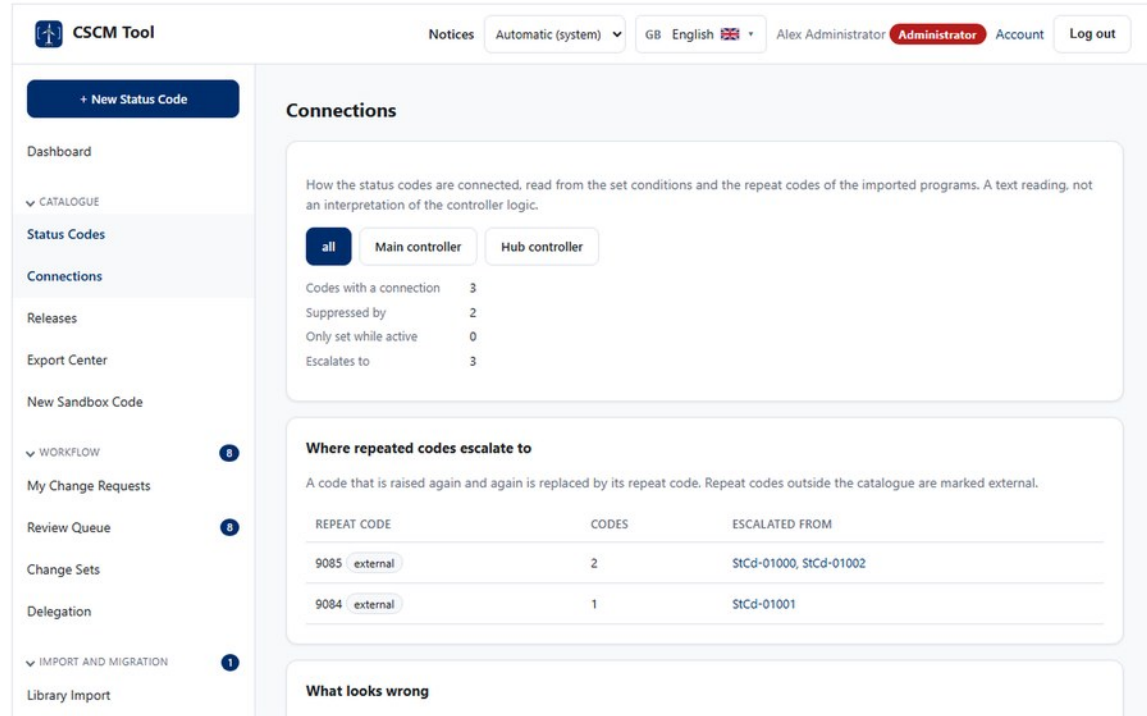
2 **A package for every audience**
All languages, access levels and formats in one download.

3 **A clear download rule**
Everyone downloads what is released; the working set stays internal.

# Insight for engineering




Catalogue health: how full each number range is, and what stalls.




Connections: which code suppresses, requires or escalates to which.

# Recently delivered

| | | |
|---|---|---|
| v4.14.0 | Dark mode | FR-183 |
| v4.14.1 | Visible version | FR-184 |
| v4.15.0 | Release documents as PDF | FR-185 |
| v4.16.0 | Design showcase decks | FR-186, FR-187 |
| v4.16.0 | Hand-over to the team repository | FR-188 |
| v4.16.1 | No assumed values in the .st import | FR-189, FR-190 |

# Governed, traceable, ready for the first migration

- Every code created, reviewed, approved and released in one tool
- Bulk work for the migration, kept honest by a sample inspection
- Languages, accessibility and a dark theme from the start

